Abdullah Sameh

DEVOPS ENGINEER

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SUMMARY

Computer Science student with strong hands-on DevOps experience spanning Kubernetes orchestration, GitOps delivery, cloud infrastructure automation, and full-stack containerization. Built and deployed production-grade systems including a live federated social media platform on Kubernetes. Focused on cloud-agnostic, maintainable architectures and automation-first practices.

TECHNICAL SKILLS

Cloud	AWS (EC2, S3, IAM, VPC, EKS, CloudFront, CodeDeploy)	**Observability**	LGTM (Grafana, Loki, Tempo, Mimir), Alloy, Prometheus, OpenTelemetry
Containers & Orchestration	Docker, Kubernetes, Helm, Kustomize	**IaC**	Terraform, Packer
CI/CD & GitOps	GitHub Actions, ArgoCD, CodeDeploy	**Scripting**	Bash, Python

PROJECTS

Mastodon — Production Kubernetes Deployment (Federated Social Media Platform)

github.com/mastodon-SRE

- Building a Kubernetes deployment for a live federated social media platform (**Mastodon**), designed to run on **any online/cloud cluster** — AWS, Civo, or otherwise — with **autoscaling** across web, streaming, and Sidekiq, plus a companion **fediscoverer** service that scales the same way.
- Self-hosted, highly-available data layer: **PostgreSQL via CloudNativePG** (primary + replica), **Redis via DandyDeveloper's redis-ha chart** (Sentinel + HAProxy) for cache and Sidekiq queues, and **Elasticsearch via ECK** — no managed DB, cache, or search service dependency.
- Storage relies on whatever CSI the target cluster provides, and ingress is handled by **Traefik** — keeping the entire repo portable across cloud providers rather than tied to any one vendor's tooling.
- Cost-optimized node architecture: stateless app tier on **Spot instances**; stateful data services (PostgreSQL, Redis, Elasticsearch) on **On-Demand nodes**.
- Full observability via a self-hosted **LGTM stack** (Grafana, Loki, Tempo, Mimir/Prometheus) running on the cluster, using **Cloudflare R2** for object storage for Mastodon media and LGTM monitoring data, plus a basic **ChatOps** setup for day-to-day operational tasks and alerts.
- Two independent CI/CD pipelines via **GitHub Actions** — one for the app, one for infrastructure — both synced through **ArgoCD** GitOps, with TLS via cert-manager and secrets via Sealed Secrets.

MERN E-Commerce — AWS 3-Tier Production Architecture

github.com/Passw0rd404/mern-ecommerce-DevOps-version

- Deployed a full-stack MERN app on AWS using a 3-tier architecture: **CloudFront → NLB → EC2 Auto Scaling Group** in a private subnet, with **ElastiCache** and **MongoDB Atlas** via PrivateLink.
- Two separate pipelines: frontend via S3 + CloudFront invalidation; backend deployed in-place one instance at a time using **AWS CodeDeploy**. Infrastructure provisioned with **Terraform**, AMIs baked with **Packer**.
- Integrated **Grafana Cloud** over **AWS PrivateLink**: Grafana Cloud collects AWS metrics through its CloudWatch data-source integration, while **Grafana Alloy** on each EC2 instance forwards telemetry through the private connection.
- Managed **MongoDB Atlas with Terraform**, authenticating Terraform with a standard MongoDB Atlas API token; monitored database health using the **Atlas monitoring dashboard**. Secured AWS deployment authentication with OIDC and added IaC scanning (TFLint, Terrascan, Checkov) and code-quality gates (ESLint, Semgrep, OpenSSF Scorecard).

HiveBox — IoT Temperature API (Built from Scratch)

github.com/Passw0rd404/HiveBox

- Wrote a **FastAPI** application from scratch aggregating real-time temperature data from global IoT sensors (OpenSenseMap API); containerized and deployed locally on **kind**.
- Integrated **Valkey** caching, Prometheus metrics, and a Grafana dashboard — full local observability stack without any cloud dependency.

HealthAI Labs — AI-Powered Health Platform (Built from Scratch)

github.com/co-op-projects/HealthAI-Labs

- Built an AI health application from scratch integrating LLM APIs; containerized with **Docker Compose** — first experience shipping an AI-integrated service end to end.
- Practiced multi-service orchestration, environment variable management, and team collaboration with Git branching and pull request workflows.

EDUCATION

Bachelor's in Computer Science, Faculty of Artificial Intelligence

Expected 2028

Minoufia University — Egypt · GPA: 3.0

CERTIFICATIONS

AWS Certified Solutions Architect – Associate	Amazon Web Services	View credential ↗
AWS Certified Cloud Practitioner	Amazon Web Services	View credential ↗
AWS DevOps — 90%	Cloud Native Base Camp	View credential ↗
McKinsey Forward Program	McKinsey & Company	View credential ↗